

The Arena Sakharov Presupposed: How Event Density Supplies the Medium Induced Gravity Needs

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Preamble — what this article does and does not claim

1. **This grounds the two things Sakharov's induced gravity *presupposes*, not a from-nothing derivation.** Sakharov (1967; modern review Visser, gr-qc/0204062) showed that the one-loop effective action of matter fields on a **given** Lorentzian manifold automatically contains the cosmological constant, the Einstein-Hilbert action, and curvature-squared terms. Two things are assumed and left to a deeper theory: the **manifold** itself (“assume you are given a Lorentzian manifold”) and the **cutoff** that regulates the divergent vacuum sum. Event Density (ED) supplies both. This article arranges ED's standing results against Visser's account; it does not re-derive Sakharov's loop.
2. **ED supplies the arena at *acoustic-metric* tier.** ED derives an emergent Lorentzian metric ($g \sim 1/b$, Paper_GR-I) from its substrate — a genuine “reasonably large, reasonably flat arena,” and with a *real medium* underneath it (the thing analog-gravity models lack). But this is the kinematic/acoustic metric; the **full curvature-emergence program** (deriving the metric's curvature dynamics and its coupling coefficients from the raw participation graph) is the standing **open** bridge under every geometric result in the corpus, and is not closed here.
3. **ED does not carry out the one-loop induction of R on its own manifold.** Sakharov's mechanism (the heat-kernel / Seeley-DeWitt expansion of the matter determinant) is *his*; ED's emergent matter would realize it on ED's emergent arena, but that computation is not done here. What ED contributes is the arena and the cutoff the mechanism requires.
4. **ED inherits Sakharov's cosmological-constant embarrassment.** A grain-cutoff vacuum energy overshoots the observed Λ by tens of orders of magnitude in ED (Paper_038_5, $\sim 10^{60}$) just as Sakharov's κ^4 term overshoots by $\sim 10^{122}$. ED does **not** solve the cosmological-constant problem; this is flagged, not papered over.
5. **ED's G is substrate-native, not Sakharov's spectrum formula.** ED's metric stiffness comes from the substrate's own always-on connectivity, a *different source* than integrating out a matter loop, so ED does not reproduce Sakharov's “ G^{-1} from the particle spectrum $\text{str}[k_1]$.” ED gives an induced-*strength* G from the grain; Sakharov's matter loop would be an additional contribution on top.
6. **ED's results are conditional derivations from thirteen posited primitives**, never derivations from nothing, and the framework is not experimentally confirmed. Tiers are marked per claim.

1. What Event Density is, and the family this belongs to

Event Density is a discrete, relational substrate: a participation graph whose channels carry a bandwidth $b \geq 0$ and a phase, with one process primitive, **commitment** (P11), the irreversible act that writes the arrow of time into the law. Gravity is emergent: the metric is the *reading* of the bandwidth field's connectivity ($g \sim 1/b$, Paper_GR-I), and Newton's constant is a conversion factor tied to the grain ($G = c^3 \ell_P^2 / \hbar$, the constants ledger, Paper_027).

This is the fourth paper in the *microscopic-completions* family, each of which takes a respected “gravity from a deeper principle” result and shows ED supplies the microscopic input its authors flagged as owed to a deeper theory: Jacobson's equation of state (Paper_GravityAsEquationOfState), Padmanabhan's horizon equipartition (Paper_GravityAsHorizonEquipartition), the khronometric Clausius derivation (Paper_KhronometricEquationOfState), and now Sakharov's induced gravity. The through-line is the same asset in different clothes: a Planck-grain, participation-based substrate beneath a presupposed geometry.

2. Sakharov's move, and the two things he presupposes

Following Visser: Einstein gravity has two ingredients — (1) the pseudo-Riemannian geometry (from the equivalence principle) and (2) the field equations for the Ricci tensor. **Sakharov addresses only (2)**. He assumes you are handed a Lorentzian manifold, makes no assumption about its dynamics, does *not* quantize the geometry, and quantizes everything else. Then the one-loop matter effective action is *guaranteed* to contain

$$\int d^4x \sqrt{-g} \{c_0 + c_1 R(g) + c_2 (R^2)\},$$

which is, term for term, the cosmological constant, the Einstein-Hilbert action $R/16\pi G$, and higher-curvature corrections. Using the heat-kernel expansion with a short-distance cutoff κ (dimensionally $[s] = \kappa^{-2}$), the divergences organize by power:

- a κ^4 divergence induces the cosmological constant, $\Lambda \sim \kappa^4$;
- a κ^2 divergence induces Newton's constant, $\frac{1}{G} \approx -\frac{1}{2\pi} \mathrm{str}[k_1]$, (Visser eq. 29), so **G^{-1} scales as the cutoff squared, weighted by the particle spectrum** through the Seeley-DeWitt coefficient k_1 ;
- logarithmic divergences induce the curvature-squared couplings.

The central lesson Sakharov draws, and the one this article builds on, is **not** the particle-spectrum formula but the deeper structural fact Visser states in his conclusion:

“Deriving the inverse square law, and all of Einstein gravity, is in fact automatic once you have demonstrated the existence of Lorentzian manifolds — reasonably large reasonably flat arenas on which to set up a low-energy QFT.”

In other words, Sakharov **reduces the problem of gravity's dynamics to the problem of the arena's existence**. Given a manifold, the dynamics comes for free at one loop. The two things left owed to a deeper theory are therefore precisely: **the manifold** (Sakharov assumes it) and **the cutoff** (Sakharov leaves it to Planck-scale physics). ED supplies both.

3. ED supplies the arena — from a real substrate, not by assumption

Sakharov’s manifold “flaps in the breeze”: it is posited, an external classical background with no account of where it came from. ED’s is not posited. The emergent metric is the *reading* of the substrate’s bandwidth connectivity:

- **The metric form is derived.** Paper_GR-I derives the emergent line element with $g_{ij} \sim b^{-1}$, lapse $N^2 \sim b$, and the Schwarzschild relation $g_{00} g_{rr} = -1$ on the Einstein branch. This is a genuine emergent **Lorentzian** geometry — a “reasonably large, reasonably flat arena” for a low-energy effective theory, exactly what Sakharov’s construction needs.
- **And it has a real medium.** Where analog-gravity models supply an acoustic metric but hedge it as an *analogy* (there is no claimed real medium under real spacetime), ED supplies the medium: the discrete participation substrate. This is the ontological upgrade Sakharov’s presupposed manifold lacks. ED’s arena is not “assume a manifold”; it is “the manifold is what the substrate’s connectivity looks like when coarse-grained.” (That ED’s emergent metric is an acoustic metric with a real medium under it is the thesis of the analog-gravity completion, target #4 in this folder’s README; it is invoked here as the ontological upgrade to Sakharov’s arena, and established there, not here.)

Honest tier. The metric is derived at *weak field* (GR-I, khronometric, coinciding with general relativity there), which is precisely the tier Sakharov’s *local* construction actually uses — a locally-flat patch to host a QFT. The metric *form* and the gravitational effective-action *form* (khronometric, one open stiffness coefficient) are derived; deriving that coefficient and the full $\sqrt{-g}$, $\mathbb{R}$ dynamics from the raw graph is the open **curvature-emergence** program. The corpus establishes the (ii)-grade emergent metric as achievable and an (i)-grade first-principles derivation of the full curved Einstein dynamics as *not* achievable under current axioms (ED_Geometry_Emergence_Scoping), and flags curvature emergence open under every geometric result (Paper_AreaLawIsTheEdgeCount). So ED supplies the arena Sakharov needs at the tier his local construction uses, while the global, fully-dynamical curvature account remains ED’s open frontier.

4. ED supplies the cutoff — and more: a substrate-native rigidity of gravitational strength

Sakharov’s induced $G^{-1} \sim \kappa^2$ makes the load-bearing input a bare cutoff κ , handed to Planck-scale physics. ED **has** that cutoff natively: the grain $\ell_{ED} = \ell_P$ (P08), $\kappa = 1/\ell_P$. Substituting gives ED’s constants-ledger value

$$G = \frac{c^3 \ell_P^2}{\hbar},$$

the “first shadow” — a conversion factor at the seam where mass meets geometry, absent from the entire Standard Model, which the ledger reads as a tell that gravity is not fundamental (Constants_Ledger_and_G_Dimension, Paper_027). The dimensional content matches Sakharov exactly, $G^{-1} \sim \kappa^2 \sim 1/\ell_P^2$ — though this is a *cutoff identification* (ED’s grain = Sakharov’s κ), not a derivation of G ’s numerical value, which is inherited (Paper_027).

But ED supplies *more* than the cutoff, and here is the one genuinely new observation of this paper, stated carefully. ED already **derives** a metric stiffness independent of any matter loop:

$$M_P^2 \sim \frac{s_{02}}{\ell_P^2},$$

the coefficient of the spatial-curvature energy, set by the **always-on** P02 reciprocal edge-sharing of the substrate — Planck-dense, present at every adjacency every tick whether or not anything commits (Paper_GR-IV §4). This is dimensionally and conceptually Sakharov’s “metric elasticity” $G^{-1} \sim \kappa^2$, but its **source is different**: it is the substrate’s own standing connectivity, not the vacuum fluctuations of matter integrated out at one loop. So:

In Sakharov’s picture, gravity’s rigidity is **induced by matter** (no matter loop, no G). In ED, gravity’s rigidity is the substrate’s **own standing elasticity** — pre-matter, kinematic — and Sakharov’s matter loop would be an *additional* contribution riding on top. ED does not need the particle spectrum to have gravity; it needs the grain and the substrate’s sharing.

This is a real distinction, not a re-labeling: ED’s metric rigidity is a property of the substrate’s standing connectivity, so ED obtains a G of the right *strength* ($\sim \kappa^2 \sim 1/\ell_P^2$) from the grain and the sharing fraction alone, without needing a matter spectrum to generate it. Note the terminology: ED’s rigidity is *not* “induced” in Sakharov’s sense — his term of art for a coupling generated by the matter loop — precisely because it is standing / always-on rather than loop-generated. Sakharov’s matter-loop contribution would then *add* to this substrate-native rigidity rather than being its sole source. (This is a claim about the *source* of the rigidity, not about its spatial variation: like Newton’s constant in either picture, ED’s M_P^2 is essentially constant, set by the grain, not tracking local matter density.)

5. The synthesis, and its honest boundary

Put the pieces together with Sakharov’s own reduction. Sakharov: *given the arena, the dynamics is automatic*. ED: *here is the arena (emergent metric, real medium) and the cutoff (grain), and moreover a substrate-native rigidity*. The two views compose:

- ED builds the arena the substrate coarse-grains into (§3);
- ED supplies the cutoff and a derived induced-strength G (§4);
- Sakharov’s one-loop mechanism — which ED’s *emergent* matter would realize on that arena — then generates the *form* of the Einstein-Hilbert action automatically (alongside the induced Λ and R^2 terms), as it does on any manifold.

And this dovetails with the rest of the family: ED already reaches the Einstein equation two other independent ways — its dynamical bandwidth rule (Paper_GR-III) and Jacobson’s clean-equilibrium equation of state (Paper_GravityAsEquationOfState, Paper_KhronometricEquationOfState). Sakharov adds a third lens: the same Einstein dynamics is what the matter loop induces on ED’s emergent arena. (The induced-gravity program even has its own horizon-entropy branch — Frolov-Fursaev’s induced-gravity black-hole entropy — that sits directly alongside ED’s Jacobson/Padmanabhan horizon results; the microscopic degrees of freedom that induced gravity attributes to heavy matter fields would plausibly correspond, in ED, to the severed participation channels — an analogy worth flagging, not a computation done here.)

The boundary, stated plainly. What this paper does *not* deliver:

- the one-loop induction of R computed on ED’s *own* emergent manifold with ED’s *own* emergent-matter spectrum (that is a real calculation, not attempted; Sakharov’s result guarantees its *form*, not its coefficients in ED);

- the closed curvature-emergence bridge (the open coefficient of ED’s gravitational effective action; §3);
- a solution to the cosmological-constant problem. On the contrary, ED reproduces Sakharov’s *embarrassment* faithfully: the naive grain/ V_1 -cutoff estimate of the induced vacuum energy overshoots the observed Λ by $\sim 10^{60}$ (Paper_038_5 §3.4, preserved on the record as a failed estimate), just as Sakharov’s κ^4 term overshoots by $\sim 10^{122}$. ED’s Λ has a structural form ($\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$, tied to the cosmic horizon; Paper_KM-II) but the smallness is inherited, not derived. **Induced gravity’s worst problem is inherited, not fixed, by ED.**

6. Honest tiers

- **Derived (ED):** the emergent metric form $g \sim 1/b$ (Paper_GR-I, conditional/khronometric); the substrate-native metric stiffness $M_P^2 \sim s_{\{02\}}/\ell_P^2$ (Paper_GR-IV §4); $G = c^3\ell_P^2/\hbar$ (constants ledger).
- **Grounded / assembled (literature + ED):** the identification of ED’s grain with Sakharov’s cutoff; the reading of ED’s arena as the manifold Sakharov presupposes; the synthesis of §5. Sakharov’s one-loop result is Visser’s, cited, not re-derived.
- **Acoustic-tier / open:** ED’s arena is the kinematic/acoustic metric; full curvature emergence (coefficients from the graph) is open (ED_Geometry_Emergence_Scoping, Paper_AreaLawIsTheEdgeCount).
- **Not attempted:** the one-loop induction of R on ED’s emergent manifold with ED’s emergent matter.
- **Inherited / unsolved:** the cosmological-constant smallness ($\sim 10^{60}$ overshoot, Paper_038_5); the value of the grain.

7. Position statement

Sakharov’s induced gravity is often read as “gravity comes from the vacuum fluctuations of matter.” Visser’s sharper reading is that Sakharov *reduced* gravity to the existence of an arena: given a Lorentzian manifold to host a quantum field theory, all of Einstein gravity is automatic at one loop, and the only things owed to a deeper theory are the manifold and the cutoff. Event Density is a candidate for that deeper theory, and it supplies exactly those two owed things: it **builds** the arena as the coarse-grained reading of a discrete participation substrate — an emergent Lorentzian metric with, for once, a *real medium* beneath it — and it **is** the cutoff, the Planck grain that fixes $G = c^3\ell_P^2/\hbar$. It even supplies a substrate-native gravitational rigidity of the right strength ($\sim \kappa^2$), sourced by the substrate’s standing connectivity rather than by a matter loop, so that ED does not wait on a particle spectrum to set G . What ED does not yet do is compute the induced action on its own arena, close its own curvature-emergence bridge, or cure the cosmological-constant problem it faithfully inherits. The claim is therefore precise and bounded: **induced gravity presupposes a medium and a cutoff; Event Density supplies both, and gives the medium a real substrate — the arena Sakharov assumed, now built rather than borrowed.** None of this is offered as confirmed physics; it is offered as a grounding of the inputs Sakharov’s construction was waiting for.

Cross-references

- Sakharov, A. D., *Sov. Phys. Dokl.* **12**, 1040 (1968); Visser, M., “Sakharov’s induced gravity: a modern perspective,” *Mod. Phys. Lett. A* **17**, 977 (2002), gr-qc/0204062 (read in full: the one-

loop effective action eqs. 1–2; $1/G \sim \text{str}[k_1]\kappa^2$ eq. 29; the “reasonably large reasonably flat arenas” conclusion, §9).

- **ED arena / curvature emergence:** physics-papers/gravity/Paper_GR-I_WeakFieldEinsteinMetric.md ($g \sim 1/b$); event-density/theory/ED_Geometry_Emergence_Scoping.md (§3F Sakharov, “presupposes a metric”; (ii)-grade achievable, (i)-grade not under current axioms); physics-papers/substrate-evaluation/Paper_AreaLawIsTheEdgeCount.md (curvature emergence flagged open).
- **ED cutoff / stiffness / G:** physics-papers/gravity/Paper_GR-IV_ArrowsAlibi.md §4 ($M_P^2 \sim s_{\{0\}}/\ell_P^2$, always-on P02 sharing); physics-papers/constants-inherited/Constants_Ledger_and_G_Dimension.md + Paper_027_Newtons_G.md ($G = c^3\ell_P^2/\hbar$, the first shadow).
- **ED Λ (the inherited problem):** physics-papers/gravity/Paper_038_5_Lambda_V1_Cosmological.md ($\sim 10^{60}$ overshoot, on the record); Paper_KM-II_KhrononCosmology.md §7 ($\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$).
- **Family companions:** Paper_GravityAsEquationOfState.md (Jacobson), Paper_GravityAsHorizonEquipartition.md (Padmanabhan), Paper_KhronometricEquationOfState.md (khronometric Clausius); README.md.
- **Induced-gravity horizon entropy (adjacent):** Frolov–Fursaev, e.g. hep-th/9703178, gr-qc/9907046 (black-hole entropy in Sakharov’s induced gravity).